

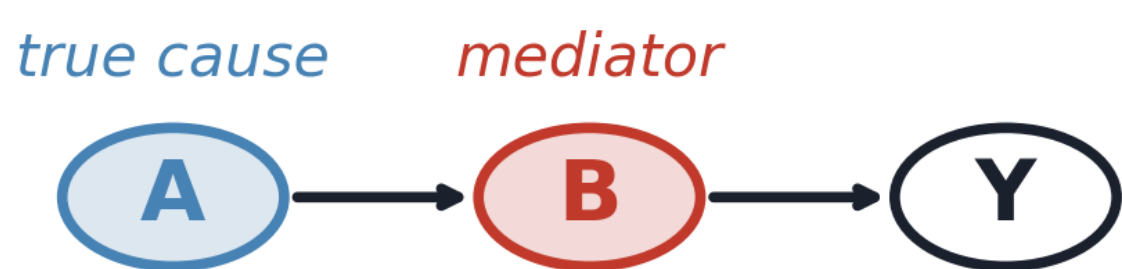
SHAP has short memory.

Causal SHAP remembers the DAG.

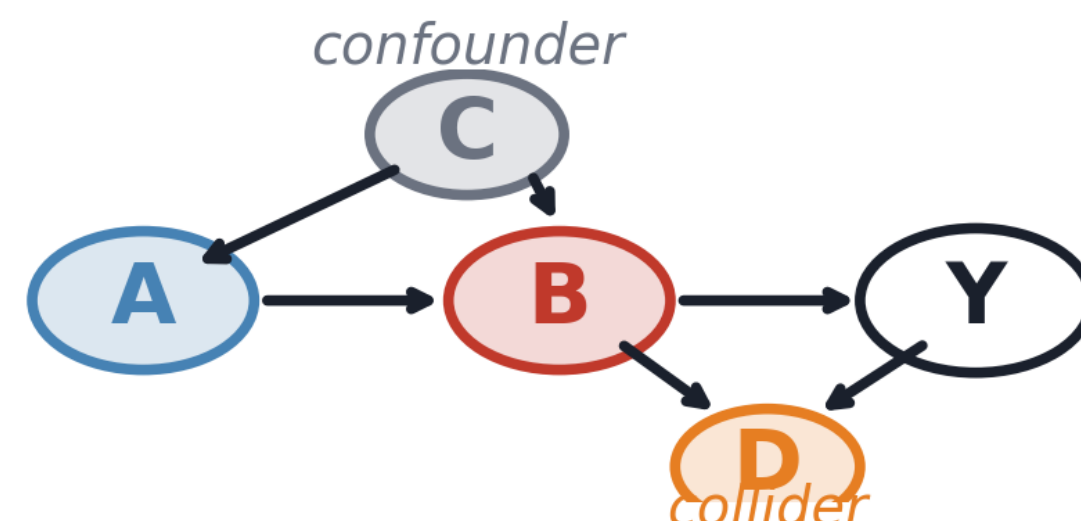
Expert-Augmented Causal SHAP: Recovering DAG-Consistent Feature Importance
via Iterative Causal Discovery and Domain Knowledge

Lexi Pasi¹ · Aimee Harrison² · Justin Ross³ · Jenny Alderden⁴ · Andy Wilson^{1,2}

¹ Lucidity Sciences ² Tao of RWD / Navidence ³ University of Utah ⁴ Boise State University



SHAP thinks:
B → Y
(misses A)



SHAP thinks:
B → Y, D → Y
(confounder leaks, collider flips sign)

WHY THIS MATTERS

SHAP is the dominant tool for explaining ML models in health research.

When features have causal structure, SHAP misattributes importance:

- Mediators absorb credit from upstream causes (topological artifact)
- Downstream interventions appear as risk factors
- Colliders, when conditioned on, create spurious attributions

Causal Shapley values (Heskes 2020; Janzing 2020) correct this by computing attributions under interventional distributions — but require a known DAG.

Fully automated causal discovery leaves edges ambiguous. Our fix: put the expert in the loop.

THE WORKFLOW

The workflow



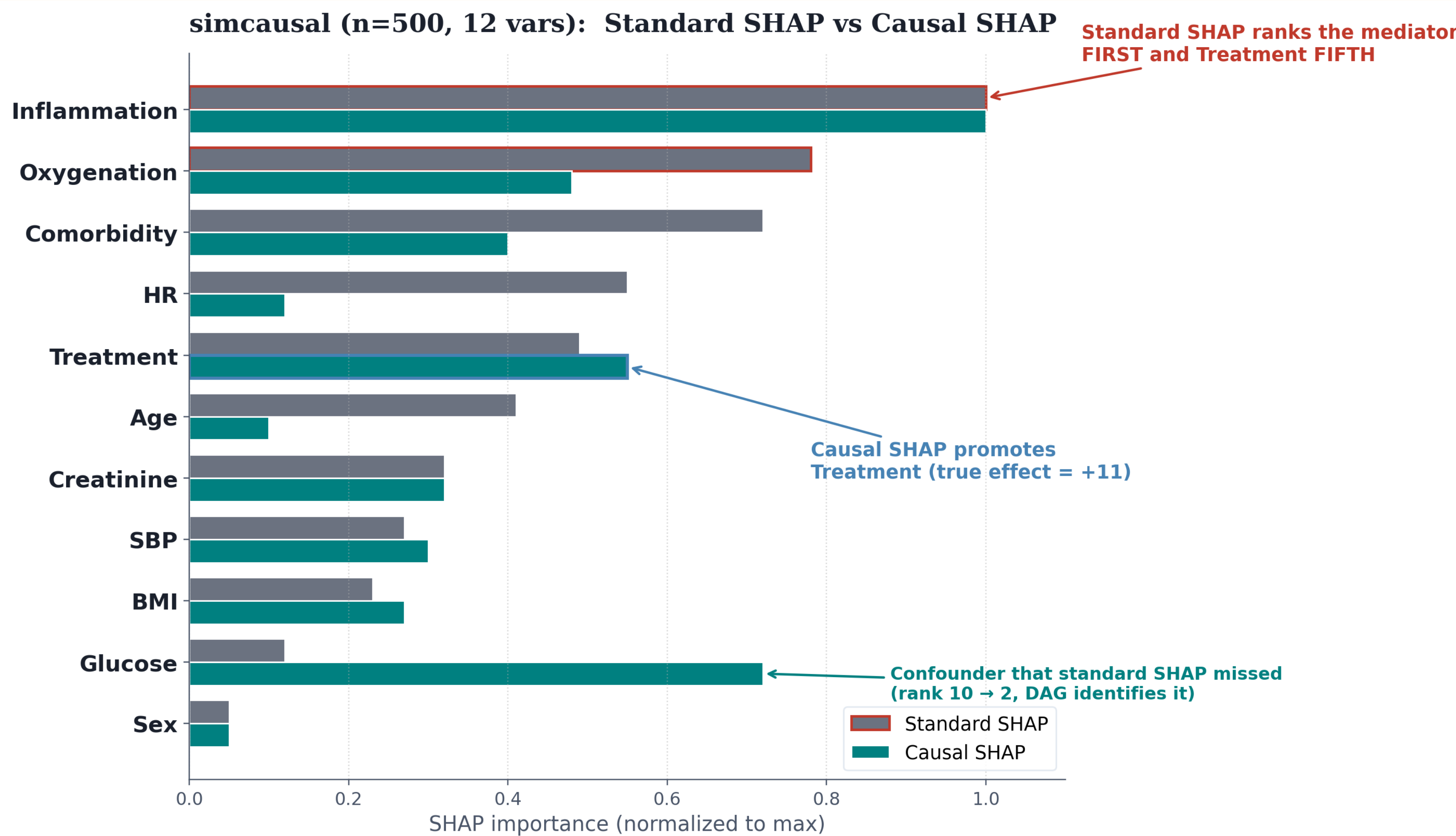
Two attribution approaches:

- DAG-constrained SHAP — restrict permutations to topological orderings (asymmetric Shapley; Frye et al. 2020)
- Adjustment-set SHAP — apply standard SHAP only to DAG-identified confounders (Shrier-Platt)

Python Shiny application supports:

- Three discovery algorithms (PC, DirectLINGAM, GES via causal-learn)
- Hard-enforced required / forbidden edges
- Immediate re-computation of SHAP as the DAG is edited
- Shrier-Platt adjustment-set export and dagitty.net integration

RESULTS · simcausal



WHEN DOES IT MATTER?



Worth the effort when:

- ✓ Treatment acts primarily through mediators
- ✓ Strong confounding (treatment-by-indication)
- ✓ Colliders included in the feature set

Ties with standard SHAP when:

- ✗ Treatment has a strong direct effect
- ✗ Features are weakly correlated
- ✗ Small graph, few structural paths

MIMIC-IV APPLICATION

n = 22,717 ICU patients · 33 variables · outcome: hospital-acquired pressure injury

Vasopressors go to the sickest patients — hemodynamic instability, sepsis, prolonged immobility.

Standard SHAP ranks vasopressor use near the top. That reflects severity, not a causal effect of the drug.

Expert DAG (40 directed edges; demographics → labs → vitals → interventions → outcome) routes credit to upstream severity drivers.

CONCLUSIONS

- Causal SHAP corrects the topological artifact — rearranges rankings, deflates mediators.
- Improvement over standard SHAP depends on structure: biggest when effects are mediated or confounded.
- No existing tool formalizes the iterative expert-in-the-loop refinement from CPDAG to resolved DAG to causal attribution.
- Our application lets epidemiologists observe how DAG edits affect attributions in real time.

KEY REFERENCES

Heskes, Bucur & Claassen (2020). Causal Shapley values. NeurIPS.
Janzing, Minorics & Blöbaum (2020). Feature relevance: A causal problem. AISTATS.
Frye, Rowat & Feige (2020). Asymmetric Shapley values. NeurIPS.
Wang, Wiens & Lundberg (2021). Shapley Flow. AISTATS.
Shrier & Platt (2008). Reducing bias through DAGs. BMC Med Res Meth.
Alderden et al. (2024). XAI for pressure injury risk. AJCC.
Johnson et al. (2023). MIMIC-IV. Scientific Data.



Interactive dashboard · live demos